MINJAE LEE

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RESEARCH INTERESTS

Robotic Foundation Models and their Adaptation, Visuo-Tactile World Models, Force Prediction, Closed-Loop Control, Multi Robot System, Dynamics Learning

EDUCATION

Georgia Institute of Technology — Atlanta, GA, USA
M.S. in Robotics (Now serving as a Teaching Assistant for Computer Vision) — *Aug. 2026 – May 2028 (Expected)*
Exchange Student, Computer Science (GPA: 4.00/4.00) — *Jan. 2022 – May 2022*

Seoul National University — Seoul, Republic of Korea
B.S. in Computer Science and Engineering (GPA: 4.12/4.30 | Rank: 4/110) — *Mar. 2019 – Jul. 2026*

Daegu Science High School — Daegu, Republic of Korea
Science and Mathematics High School for Gifted Students (My focus: Physics & Computer Science) — *Mar. 2016 – Feb. 2019*

SKILLS

Programming: Python, C/C++, Java
Software and Tools: PyTorch, TensorRT, ROS, Gazebo, Unity, Git, Linux, mjlab
Languages: English (Fluent; TOEFL 110), Korean (Native)

EXPERIENCE

Georgia Institute of Technology – LIDAR Lab — Atlanta, GA, USA
Graduate Student Researcher — *Aug. 2026 – Present*

- Developing a decentralized framework for communication-free cooperative humanoid transport using local sensing and object-centric planning.
- **Local Object Belief Estimator.** Developed a multimodal local object belief estimator that fuses vision, proprioception, and wrist force–torque histories for payload state estimation and action-conditioned wrench prediction.
- Examining how to make action models adaptive to the environment using tactile information.

SNU Machine Perception and Reasoning Lab — Seoul, Republic of Korea
Research Intern — *Sep. 2024 – Present*

- **In-Context Adaptation for VLA Policies.** Investigated few-shot adaptation from retrieved demonstrations across VLA architectures in cross-suite transfer tasks on LIBERO. Found that existing VLA/VLM ICL methods fail to effectively utilize demo context. They relied heavily on demonstration actions without adequately relating them to the corresponding images and instructions, failing to adapt actions appropriately when the current observation differs from the corresponding observation in the demonstration.
  Achieved an average success rate of **56.3% versus 44.7% for RICL**, the strongest baseline in our cross-suite setup, by integrating **NVIDIA GR00T N1.7** with a novel architecture that attends to the relationships between demo action and demo image-instruction pairs and using a causal masking for variable-shot learning.
- **Memory for Long-Horizon VLA Manipulation.** Developed a progress-aware episodic-memory pipeline based on GR00T N1.5 for partially observable manipulation by compressing interaction history into non-local keyframes recalled at each step, and training memory use with reinforcement learning. Outperformed vanilla ContextVLA (base architecture) on the target tasks.
- **Robotic Grasping Affordance Detection.** Designed and implemented zero-shot inference and visual segmentation of safe grasping regions for both human operators and collaborative robots using LLMs and image-generation models.

Turing — Seoul, Republic of Korea
AI Researcher & Engineer — *Jul. 2022 – Aug. 2024*

- Built controllable models of student ability and behavior using domain-expert knowledge, combining high predictive accuracy with model behavior consistent with human intuition.
- Devised an LLM-based approach to flipped learning in mathematics education and designed a pipeline to implement it, leading **OpenAI** to select Turing as **one of its 10 partner companies in Korea**.
- Incorporated expert knowledge into AI models for mathematical reasoning and deployed the models in the company's product.

Georgia Institute of Technology – DCSL Lab Atlanta, GA, USA
Undergraduate Student Researcher, ML subteam of RC-VIP *Jan. 2022 – May 2022*

- Developed a learned vehicle-dynamics model for predicting future trajectories.
- Reduced mean absolute error (MAE) by 90% relative to the lab's previous model over a one-second prediction horizon.

Artificial Intelligence Institute of Seoul National University Seoul, Republic of Korea
Research Intern *Jun. 2021 – Sep. 2021*

- Developed an AI model to characterize e-commerce shoppers and predict their behavior.

PUBLICATIONS

[J01] Kwangho Lee, Youngdo Kim*, Youngsi Kim*, Juho Kim*, **Minjae Lee***, Joonho Kong. (2018). "Approximate processing hardware design and implementation of exponential function presented with Taylor series for embedded systems." *Proceedings of Symposium of the Korean Institute of Communications and Information Sciences*, pp. 38–39.

[C01] Hyeondey Kim*, Jinwoo Nam*, **Minjae Lee***, Yun Jegal, Kyungwoo Song. (2023). "Leveraging Skill-to-Skill Supervision for Knowledge Tracing." *AAAI AI in Education (AI4ED) Workshop* (3 citations as of Nov. 2026.)

[C02] Sungyeon Park, **Minjae Lee**, Jihyuk Kang, Hahyeon Choi, Yoonah Park, Juhwan Cho, Adam Lee, Dongkyu Kim. (2024). "VLAAD: Vision and Language Assistant for Autonomous Driving." *WACV Workshop on Large Language and Vision Models for Autonomous Driving (LLVM-AD)* (99 citations as of Nov. 2026.)

[C03] Wongi Park, Jiyeon Lim, **Minjae Lee**, Myeongseok Nam, Seongjun Choi, Jungwoo Kim, William J. Beksi, Soomok Lee, Sang-Hyun Lee. (2026). "Rectifying Mask via Entropy for Distractor-Free 3DGS in Ambiguous Scenarios." *Accepted at WACV 2027*.

** Denotes equal contribution.*

PREPRINTS

[P01] Wongi Park, Jordan A. James, Myeongseok Nam, **Minjae Lee**, Soomok Lee, Sang-Hyun Lee, William J. Beksi. (2026). "Difix3D-W: Distractor-Free Few-Shot 3D Gaussian Splatting in the Wild." arXiv:2604.27422. *Under review*.

PATENT

Minjae Lee, Jinwoo Nam. (2023). "Method, Program, and Device for Quantifying Correlation Between Units." Korean Patent No. 10-2023-0075514.

HONORS & AWARDS

Kwanjeong Educational Foundation Study Abroad Scholarship — $30,000/year 2026–2028

Seoul National University Foundation
Lee Kwang Woo Global Engineering Talent Development Scholarship — $35,000 Jun. 2026

National Science & Technology Scholarship — 2-year full tuition (2-year leave to work at Turing excluded) 2021–2024

Mirae Asset Global Exchange Student Scholarship — $6,000 Feb. 2022

Special Appreciation Award — Char, Kook Heon, Dean, College of Engineering, SNU
Appreciation for dedicated efforts to foster young engineers Jul. 2021

PROJECTS

Creating Robot AI for Natural Language Task Execution — Seoul National University 2024

Adapter Module Implementation in Vision Transformer — Georgia Tech 2022

Virtual Clothing Try-On Using Computer Vision and Deep Learning — Seoul National University 2021

KNN-Based Disease Diagnosis and Prescription System — Daegu Science High School 2016

ACTIVITIES AND SOCIETIES

Member, AttentionX (Deep Learning R&D Club)
Researched interpretable decision-making for autonomous vehicles (Refer to VLAAD) 2023–2024

Mentor Team Lead, AI TECH PLAY Program 2021
Taught autonomous-vehicle development using MIT LL RACECAR, with access to the official instructor-only repository arranged by the program lead (an MIT graduate student).

Steering Committee Member, Daegu Science High School Code Jam 2017